Palaeomagnetic evidence shows Malay Peninsula was not a part of Gondwanaland

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Palaeomagnetic results from Malaya show that the Malay Peninsula lay at 15° N during the late Palaeozoic, a result incompatible with the hypothesis that it once formed a part of Gondwanaland adjacent to India or Australia. Cretaceous results suggest the Peninsula was not at that time finally welded to the Asian mainland.

DEBATE has recently centred around the proposition that parts of South-east Asia were once a part of Gondwanaland. Burton[1] tried to explain the westerly origin of the sediments of north-west Malaya by supposing, in a reconstruction of Gondwanaland, that the west coast of the Malay Peninsula lay adjacent to the east coast of India. His argument was supported by stratigraphic comparisons and the reasonably good fit of the coastlines. Ridd[2] went one step further and fitted the whole of mainland South-east Asia and the Sunda shelf region between western Australia on the east, and the east coast of India on the west. This reconstruction was subsequently accepted by Audley-Charles *et al.*[3] who incorporated it into their hypothesis of the development of eastern Indonesia. Stauffer and Gobbett[4], however, raised serious objections to such a reconstruction from palaeoclimatic and other evidence. During the late Palaeozoic both India[5] and Australia[6] suffered continental glaciation and lay in high southern latitudes[7]. By contrast the

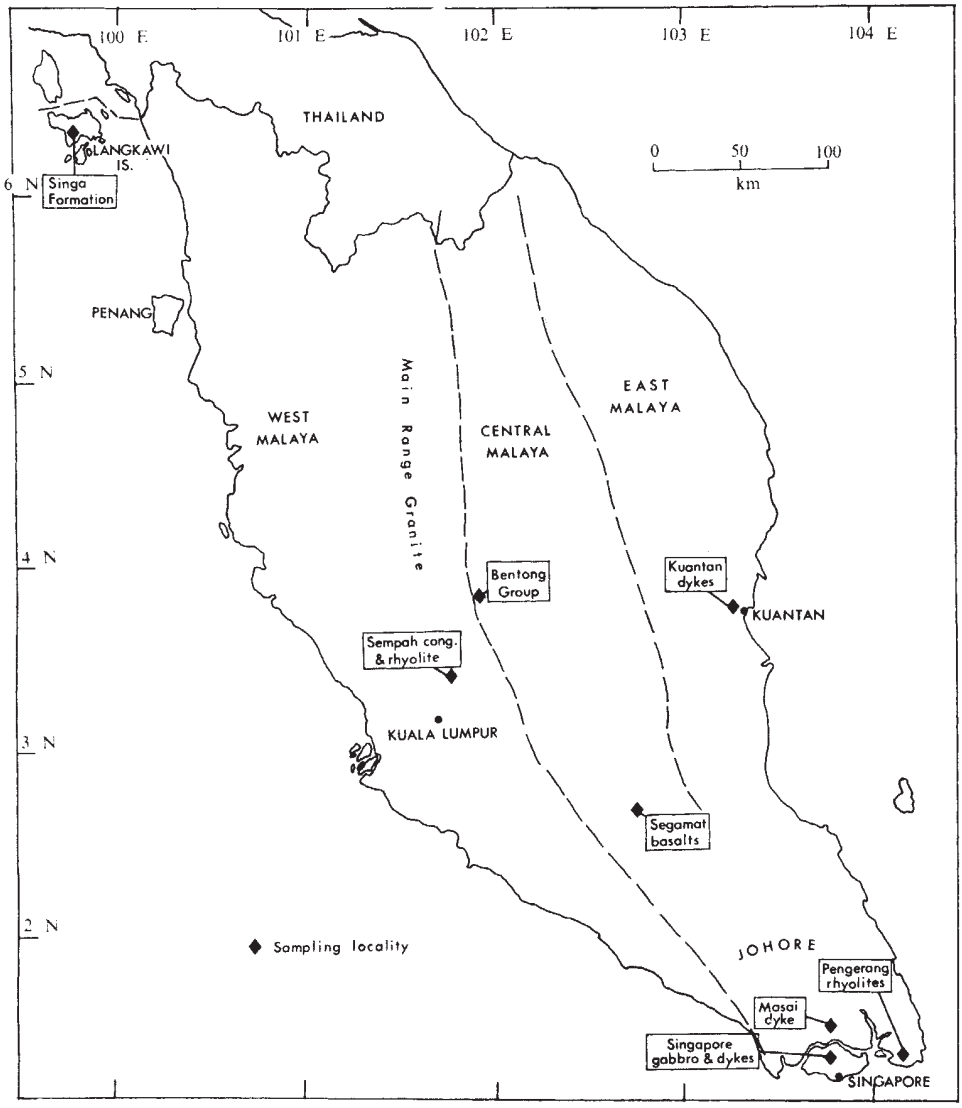


Fig. 1 Map of the Malay Peninsula showing sampling localities.

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geological record of South-east Asia generally indicates tropical or subtropical conditions during the late Palaeozoic[4]. That the Permian of South-east Asia was laid down in warm water conditions is also accepted by Ridd[8].

Both the palaeoclimatic problems and palaeogeographic reconstructions put forward can be tested using palaeomagnetic measurements. We report here the results of a reconnaissance survey of the palaeomagnetism of rocks of the Malay Peninsula. Some very preliminary results of this survey have already been reported[9].

Geology and sampling

The Malay Peninsula can be divided into three geologically distinct zones shown in Fig. 1. Since the first discovery of Lower Palaeozoic fossils by Jones[10] it has been found that Lower Palaeozoic rocks are restricted to the western zone. Recent work in Thailand[11,12] has recognised the extension of this western zone of Lower Palaeozoic rocks through peninsular Thailand into north-west Thailand and eastern Burma and northwards into Yunnan in China. This long geosynclinal belt has been referred to by Burton[11] as the Yunnan–Malay geosyncline. In the Malay Peninsula the boundary with the Lower Palaeozoic belt is approximately marked by the so-called Bentong Line[13] which includes abundant radiolarian chert, some serpentine masses, metabasites and dolerite sills. It has been suggested that this marks the former position of a late Palaeozoic subduction zone[14]. The central zone is characterised by folded clastic Triassic–Jurassic carbonate and continental deposits whilst the eastern zone is composed of predominantly deep-water marine clastic Upper Palaeozoic rocks with patches of Upper Mesozoic continental deposits. The central and eastern regions are together considered by Stauffer[15] to be an island arc formed in late Palaeozoic times and welded on to the block to the west. The most recent summary of the geology of the Malay Peninsula is given by Gobbett and Hutchison[16]. In the present study hand samples, oriented by Brunton compass, were collected from a number of sedimentary and volcanic formations and igneous intrusions ranging in age from Carboniferous to Cretaceous (Fig. 1).

Six samples of red muddy sandstone from the Bentong Group were collected between mileposts 75 and 76 on the road from Kuala Lumpur to Raub (3.8°N, 101.8°E). The dip of the beds ranges from 30° to 70°, and in this region they are overlain, apparently conformably, by fossiliferous Visean shales and sandstones[27]. The samples are thus probably Lower Carboniferous, although they could be older.

Three samples of laminated muddy sandstone were collected from the Singa Formation on the south-west part of Langkawi Island, off the north-west coast of the Malay Peninsula (6.2°N, 99.8°E). The formation covers the whole of the Carboniferous[17] and the stratigraphic position of the samples suggests they lie in the upper part of the Lower Carboniferous. The beds have an easterly dip of between 25° and 30°.

Twelve samples of reddish sandy mudstone were collected from the Sempah conglomerate together with seven samples of the overlying rhyolite in the Genting Sempah area on the Ulu Kali Road (3.4°N, 101.8°E). The rocks dip east at angles between 20° and 65°. Although inconclusive, Rb–Sr dating of the rhyolites[18] suggests they are most probably Early Permian in age with a maximum possible age of Late Carboniferous.

Ten samples were collected from rhyolites occurring at Pengerang in south-east Johore (1.4°N, 104.2°E). The lavas dip north-west at between 19° and 36°. Rb–Sr measurements on these samples suggest they are Early Triassic or Permo-Triassic in age. This is identical with the age of the south Johore and Singapore granites[18], with which they are probably associated.

Ten samples of basalt and two of red agglomerate were collected from the Segamat basalts of Negri Sembilan (2.5°N, 102.8°E) described by Grubb[19]. The lavas lie unconformably on Middle to Upper Triassic rocks. The flows appear to dip in a south-westerly direction at 15°, although this dip may be in part original. A single, whole rock, K–Ar age determination from these basalts gives an age of 62 Myr (ref. 18) but this is regarded as a minimum age in view of demonstrated argon loss from other rocks in the region[18]. The basalts are probably Cretaceous in age.

Three samples were collected from basaltic dykes near Kuantan, Pahang (3.8°N, 103.3°E) and two from a dyke at Masai, Johore (1.5°N, 103.8°E). One of the Kuantan dykes gives a K–Ar age of 110 Myr and is therefore probably Cretaceous in age[18].

One sample of gabbro and four samples, one from each of four thin dykes intruding the gabbro, were collected from two quarries on Singapore Island (1.3°N, 103.8°E). The gabbro is intruded by the Singapore granite of Early Triassic age[18] and basic dykes are not known to cut the granite. The dykes are

Table 1 Analytical data for four samples of Pengerang rhyolites (total rocks)

ANU no.	Rb (p.p.m.)	Sr (p.p.m.)	$^{87}Rb/^{86}Sr$	$^{87}Sr/^{86}Sr$
71–699	223	38	17.238	0.7692
71–700	250	35	20.584	0.7784
71–702	247	120	5.967	0.7302
71–703	224	134	4.8214	0.7273

The decay constant for ^{87}Rb used was 1.39×10^{-11} yr^{-1} with statistical parameters appropriate to the experimental precision of the instrumental procedures.
ANU, Australian National University.

Table 2 Mean directions of magnetisation observed from a reconnaissance survey of Malayan rocks

Rock unit	Age	n	D	I	R	k	α_{95}
Segamat basalts, Kuantan and Massai dykes	K	16	136	−31	15.081	16.3	9.4
Pengerang rhyolites	(Pu-)Trl	10	206	−39	9.638	24.9	9.9
Sempah conglomerate and rhyolites	(Cu-)Pl	14	212	−32	13.083	14.1	10.9
Bentong Group and Singa formation	C(l)	8	213	−16	7.725	25.4	11.1
Singapore gabbro and dykes	>Trl	5	194	+21	4.758	16.5	19.3

n, No. of samples; D, declination east of true north; I, inclination–positive downwards; R, resultant of the n unit vectors; k, precision parameter; α_{95}, radius of circle of 95% confidence about the mean.

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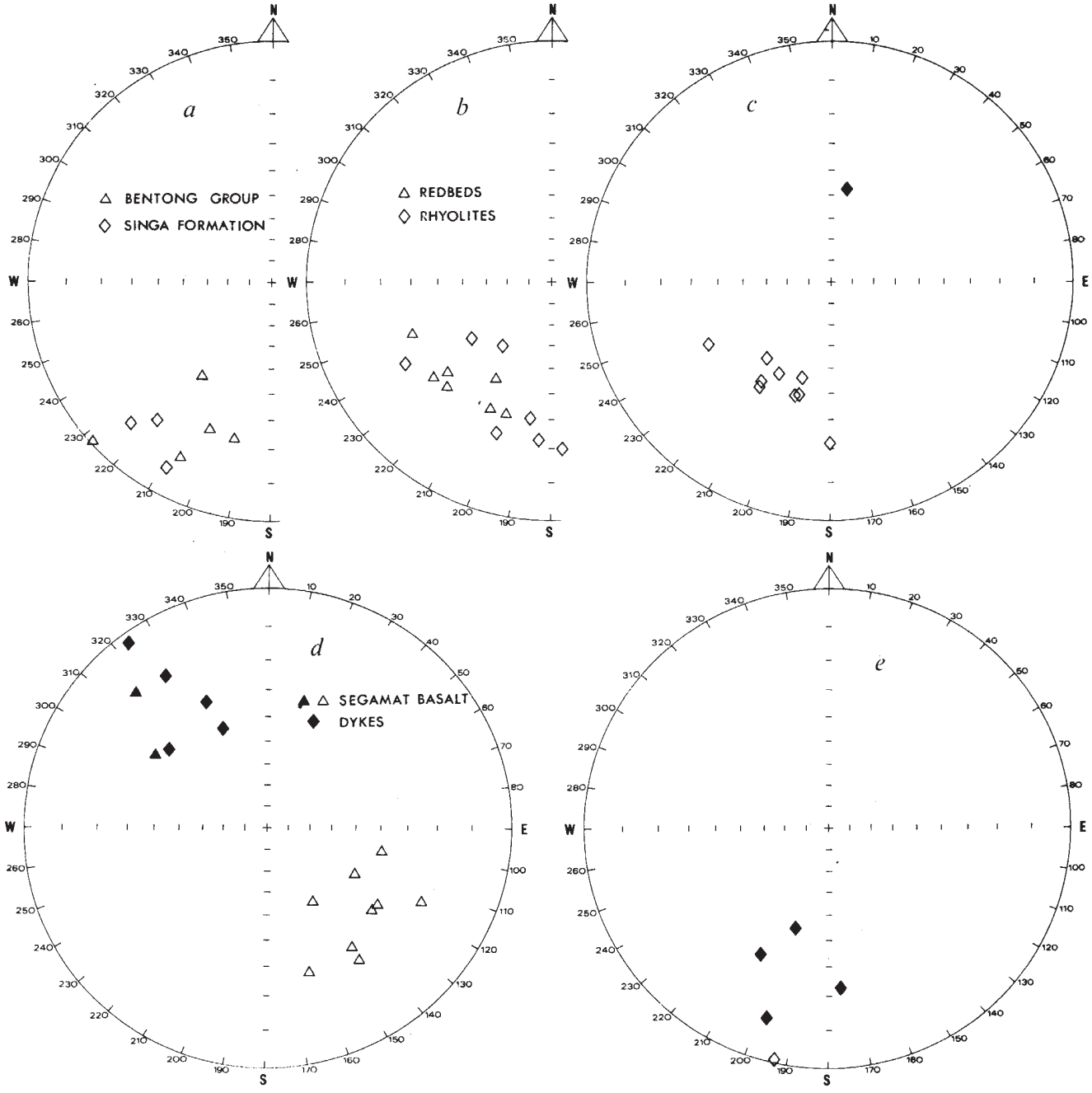


Fig. 2 Stereographic projections showing the cleaned directions of magnetisation for (*a*) Bentong Group and Singa Formation, (*b*) Sempah conglomerate and rhyolite, (*c*) Pengerang rhyolites (*d*) Segamat basalts, Kuantan and Masai dykes, (*e*) Singapore gabbro and dykes. Solid symbols on the lower hemisphere, open symbols on the upper hemisphere.

Several specimens were cored and sliced from each hand sample and the natural remanent magnetisation was measured using a PAR-SM1 magnetometer. Pilot specimens were subjected to alternating field demagnetisation in increasing fields[20]. Most of these measurements were made at the Australian National University but during the final stages of this survey some additional measurements were made in the newly established palaeomagnetic laboratory in Kuala Lumpur. Measurements were then made on a Digico magnetometer and magnetic cleaning undertaken with a Schonstedt demagnetiser. In Canberra, pilot specimens of the sedimentary samples were also subjected to stepwise thermal demagnetisation in which the specimens were heated and cooled in a magnetic field of one γ (1nT) or less[21].

probably of similar age to the gabbro which must be older than Early Triassic, but by how much is unknown, as there is no evidence which could define a maximum age.

Rb–Sr age of the Pengerang rhyolites

Four samples of Pengerang rhyolites were analysed as total rock samples at the Australian National University by the Rb–Sr method (see Table 1). The individual analyses show inadequate enrichment in radiogenic strontium to make reliable estimates of age. A statistical analysis using the least-squares method[25] with a two-error regression[26] indicates an age of 238 $\pm$ 23 Myr. The high mean square of weighted deviates (18.37) places this estimate in the category of a reconnaissance age determination, in so far as the samples probably had slightly different initial $^{87}Sr/^{86}Sr$ ratios; but the age can reliably be placed close to the Permo-Triassic boundary.

Bignell[18] obtained an isochron of 225 $\pm$ 6 Myr for a combination of three analyses of Singapore granite and two of

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Table 3 Statistical significance of the fold test applied to the four age groups using the table of McElhinny[22]

Rock unit	Age	n	Directions			Poles		
			k_b	k_a	Sig.	K_b	K_a	Sig.
Segamat basalts, Kuantan and Masai dykes	K	16	18.5	16.3	No	22.3	18.6	No
Pengerang rhyolites	(Pu-)Tr1	10	13.9	24.9	No	9.4	21.5	Yes
Sempah conglomerate and rhyolites	(Cu-)P1	14	10.9	14.1	No	6.8	13.4	Yes
Bentong Group and Singa formation	C(1)	8	9.7	25.4	Yes	17.3	37.0	No

k_b K_b are precision parameters before applying tilt corrections and k_a K_a are those after applying tilt corrections.

Johore granites, using a decay constant of 1.47×10^{-11} yr^{-1}. Converted to the 1.39 decay constant used in our analysis, this result gives a figure of 238 ± 6 Myr. Although this is also in the category of a reconnaissance result as the samples are from different localities, it is identical with that for the Pengerang rhyolites, which may provisionally be regarded as the effusive equivalent of the granites.

Palaeomagnetic results

Samples from the Bentong Group and Singa Formation responded well to thermal cleaning. Pilot specimens showed excellent stability up to the Curie Point of haematite. A small viscous component in the present field direction was removed at 300° C and on correcting for the dip of the strata, the samples group in a south-westerly direction with shallow inclinations (Fig. 2*a*). One sample was too weakly magnetised and its remanence could not be measured.

There was no difference in response of the red sediments from the Sempah conglomerate to either thermal or magnetic cleaning. Both the red sediments and rhyolite were therefore cleaned at 30 mT. Five of the twelve samples of red sediment, however, remained clustered about the present field direction, could not be cleaned and were rejected. This illustrates a problem associated with sampling red sediments in tropical countries. Even if samples are collected from road cuttings within a few years of exposure, weathering can have progressed fairly deep. In the case of red sediments it is sometimes very difficult to distinguish whether substantial deep weathering has occurred. Seven samples of red sandstone and seven rhyolites gave consistent directions after cleaning, similar to those given by the Bentong Group and Singa formations (Fig. 2*b*). All these late Palaeozoic samples are reverse magnetised.

The Pengerang rhyolites were extremely stable to alternating field demagnetisation, and blanket cleaning at 30 mT was carried out. One of the ten samples was normally magnetised and the other nine are reversed. The mean direction (Table 2, Fig. 2*c*) is very similar to that of the other late Palaeozoic groups. The basic rocks from Singapore and Johore and at Segamat and Kuantan were less stable than the rhyolites. One of the Segamat basalts proved to be unstable but the remainder of the samples could be cleaned at 20 mT. The Segamat flows are all reverse magnetised whilst the red agglomerate and the dykes from Masai and Kuantan are normally magnetised (Fig. 2*d*). The Singapore gabbro and dykes are all reverse magnetised (Fig. 2*e*).

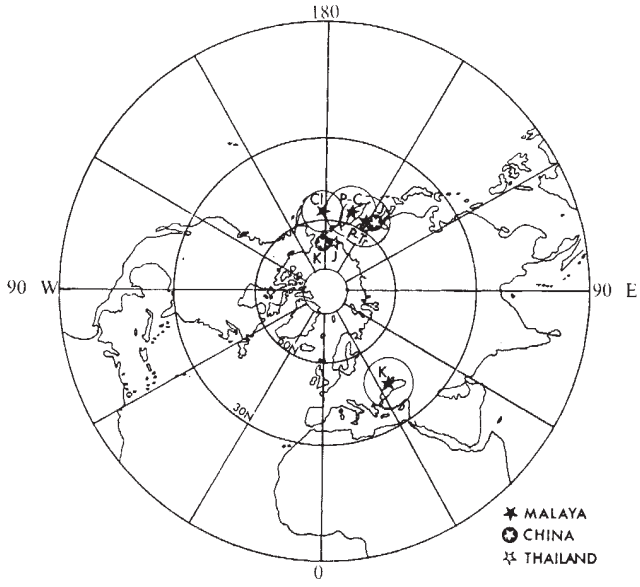


Fig. 3 Palaeomagnetic poles derived from the Malay Peninsula compared with those from China[7] and Thailand[23].

The mean direction of magnetisation for each of the groups is given in Table 2 together with the usual statistical parameters. The three late Palaeozoic means are very close to one another with only one of the 32 samples being normally magnetised.

Table 4 Palaeomagnetic pole positions calculated for the four age groups (mean of virtual geomagnetic poles)

Rock unit	Age	n	Lat. N	Long. E	K	A_{95}
Segamat basalts, Kuantan and Masai dykes	K	16	44	35	18.6	8.7
Pengerang rhyolites	(Pu-)Tr1	10	57	152	21.5	10.7
Sempah conglomerate and rhyolites	(Cu-)P1	14	55	164	13.4	11.2
Bentong Group and Singa formation	C(1)	8	57	182	37.0	9.2

n, No. of samples; K, precision parameter; A_{95}, radius of circle of 95% confidence about mean pole position.

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Because of the different attitudes of the beds sampled, substantial tilt corrections have been applied. The statistical significance of the application of the fold test to four of the groups is given in Table 3 using the tables of McElhinny[22]. In spite of the relatively small collection of samples, there is a significant fold test for either the directions or poles for three of the four groups. This argues strongly for the stability of magnetisation of the samples at least since the time of folding, generally regarded as Late Triassic. The tilt corrections are too small for the Cretaceous group to produce any positive fold test. The presence of roughly opposed directions of magnetisation (Fig. 3*d*), however, provides additional stability evidence.

Palaeomagnetic poles and palaeogeography

Palaeomagnetic pole positions for four of the age groups are listed in Table 4. The five samples from Singapore are not considered sufficient to calculate a viable pole position and, in any case, the precise age of these rocks is not known. Between the Early Carboniferous and Early Triassic there appears to be a small westerly migration of the pole but it is hardly significant (Fig. 3). In fact, the data are such that it might be valid to combine all the 32 late Palaeozoic measurements together to produce an overall pole at 56°N, 165°E with $A_{95} = 6.3°$. The distribution of these poles changes from $K = 6.1$ to $K = 17.1$ after correcting for the attitude of the beds, a change that is statistically highly significant[22].

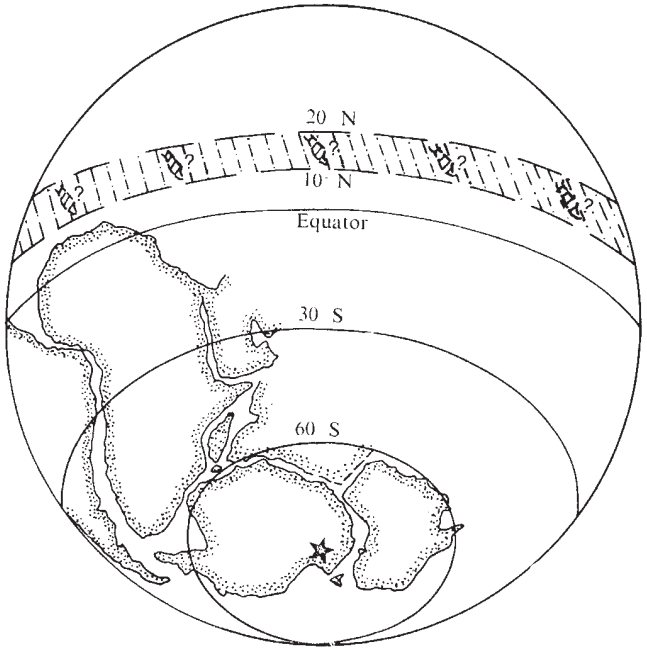


Fig. 4 Gondwanaland during the Permo-Carboniferous. The star is the mean palaeomagnetic south pole[24]. The Malay Peninsula must lie at 15°N, anywhere along the arc of a circle 105° from the pole. A 10° wide belt is shown together with some possible positions of the Peninsula.

The palaeomagnetic poles are plotted in Fig. 3 where they are compared with poles from the Jurassic of Thailand[23] and the Jurassic and Cretaceous of China[7]. These lie fairly close to the Malayan late Palaeozoic poles but are well removed from the Malayan Cretaceous pole. This would suggest that the Malay Peninsula was not firmly attached to the mainland in Cretaceous times. The magnetic inclinations observed during the late Palaeozoic and Cretaceous in Malaya are very similar (Table 2) indicating that the Peninsula continuously occupied low latitudes of about 15°N during this time. This strongly supports the contention[4] that the Malay Peninsula occupied low latitudes during the late Palaeozoic. Between the Permo-Triassic and the Cretaceous, the Peninsula underwent a clockwise rotation of 70° whilst maintaining much the same palaeolatitude.

The data for the late Palaeozoic may be used to test the hypothesis that the Malay Peninsula once formed a part of Gondwanaland. Since it is known that the late Palaeozoic is a period of predominantly reversed polarity[7] the data of Table 2 indicate that the Peninsula lay 105° away from the South Pole, that is, in the northern hemisphere at 15°N. Using the latest estimate of the Permo-Carboniferous pole with respect to Gondwanaland[24], the Malay Peninsula must lie along a circle lying 105° away from this pole (Fig. 4). The palaeomagnetic data are clearly inconsistent with a position adjacent to India or Australia in the late Palaeozoic. Since Gondwanaland did not break up until the Jurassic, it seems unlikely that the Malay Peninsula was associated with the supercontinent in the early Palaeozoic and broke away before the other segments.

From Fig. 4 it can be seen that with due allowance for possible inaccuracies inherent in the method of positioning the Peninsula and Gondwanaland, the data do not completely rule out the hypothesis of Stauffer[15] that the Malay Peninsula may have been adjacent to parts of north Africa in the Palaeozoic. Further palaeomagnetic work planned for the Lower Palaeozoic of the Malay Peninsula may throw more light on this hypothesis.

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